

VNU-HUS MAT1206E/3508: Introduction to AI

Logic Programming with PROLOG In-class Discussion

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Learn PROLOG Now!

- <https://www.let.rug.nl/bos/lpn/index.php>
- by **Patrick Blackburn, Joost Bos, and Kristina Striegnitz.**

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- **PROLOG** = **P**rogramming in **L**ogic
- PROLOG is used in many projects, primarily in *AI* and *computational linguistics*.
- We give a *short introduction*, show key concepts and strengths, and compare PROLOG with other programming languages and theorem provers.
- For a complete programming course, see [Bratko 2011]; [Clocksin and Mellish 2013].
- *Software note*: common systems are *SWI-Prolog* (<https://www.swi-prolog.org/>) and *GNU Prolog* (<https://www.gprolog.org/>). Here, ordinary examples use SWI-Prolog syntax; CLP uses GNU Prolog.
- Most modern PROLOG systems use an interpreter based on the *Warren Abstract Machine (WAM)*.
- PROLOG source code is compiled into *WAM code*, then interpreted by the WAM.
- *Performance*: up to 10 million logical inferences per second (*LIPS*) on a 1 Gigahertz PC

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Facts, Rules, and Queries



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The logical core of PROLOG uses *Horn clauses*: clauses with at most one positive literal.

Here $H, B_1, \dots, B_n$ are placeholders for atomic formulas (predicate calls).

Type	PROLOG format	Meaning
Fact	$H.$	H holds.
Rule	$H :- B_1, \dots, B_n.$	If all B_i hold, then H holds.
Query	$?- B_1, \dots, B_n.$	Ask whether all B_i hold; find values for query variables.

- In a rule, H is the *head*; the goals after $:-$ form the *body*. A fact has no body.
- $:-$ reads “if”; $,$ means “and”; $?-$ is the query prompt. A full stop $(.)$ ends each complete form.
- In clause form, the rule is $\neg B_1 \vee \dots \vee \neg B_n \vee H$: only H is positive.

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- PROLOG is a *declarative programming language*, i.e., the programmer declares *what* the program should accomplish without specifying *how* to achieve the result.
- PROLOG is based on *Horn clauses*.
- A PROLOG program consists of a *knowledge base* (*database*), which is simply a set of *facts* and *rules* about some problem domain.
 - A *knowledge base* *KB* of family relationships is coded as a PROLOG program

```
1 child(oscar, karen, frank).  
2 child(mary, karen, frank).  
3 child(eve, anne, oscar).  
4 child(henry, anne, oscar).  
5 child(isabelle, anne, oscar).  
6 child(clyde, mary, oscarb).
```

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- The execution of a PROLOG program is initiated by a *query*, which is answered by proving that the query logically follows from the facts and rules in the program.

- Example query

```
?- child(eve, anne, oscar).
```

- The query asks whether eve is a child of anne and oscar.
- The expected answer is true (because it is a fact in the knowledge base *KB*).
- How does PROLOG find the answer?
 - PROLOG tries to unify the query with the facts in the knowledge base *KB*.
 - There are six facts in the knowledge base.
 - Unification is attempted between the query and each of the complementary literals in the input data in order of occurrence. (In this example, the query unifies with the third fact.)
 - If one of the alternatives fails, this results in backtracking to the last branching point, and the next alternative is tested.

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Capitalization changes the meaning of a term in PROLOG.

Quiz

Which term contains a *variable*?

- (a) `woman(mia)`
- (b) `woman(Mia)`
- (c) `woman('Mia')`

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An *atom* is a constant term. Single quotes can make one atom out of text that would otherwise not be a simple atom.

Quiz

Which one is valid PROLOG syntax for a single *atom whose name contains a space*?

- (a) `intro ai`
- (b) `'intro ai'`
- (c) `IntroAi`

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A predicate is identified by its *functor* and *arity* (number of arguments).

Quiz

Are `loves(john, mary)` and `loves(john, mary, secret)` clauses of the same predicate?

- (a) Yes, because both use `loves`.
- (b) No, they define `loves/2` and `loves/3`.
- (c) No, because predicate names cannot repeat.

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Central Ideas of PROLOG

■ **SUCCESS (true) / FAILURE (false)**

- any computation can “succeed” or “fail”, and this is used as a ‘test’ mechanism.

■ **UNIFICATION (2-WAY MATCHING)**

- any two data items can be compared for similarity, and values can be bound to variables in order to allow a match to succeed.

■ **SEARCHING**

- the whole activity of the PROLOG system is to search through various options to find a combination that succeeds.

■ **BACKTRACKING**

- when the system fails during its search, it returns to previous choices to see if making a different choice would allow success.

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```
1 teaches(mai, ai).  
2 student(an).  
3 can_help(T,S):- teaches(T, ai), student(S).  
4 ?- can_help(mai, an).
```

Quiz

Which labeling is correct?

- (a) Line 1 is a fact, line 3 is a rule, and line 4 is a query.
- (b) Line 1 is a query, line 3 is a fact, and line 4 is a rule.
- (c) All four lines are facts.

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```
1  woman(mia).  
2  woman(jody).  
3  woman(yolanda).  
4  playsAirGuitar(jody).  
5  party.
```

```
?- woman(mia).  
?- playsAirGuitar(mia).  
?- playsAirGuitar(vincent).  
?- woman(vincent).
```

Quiz

For the above queries, which option correctly describes the expected Prolog answers?

- (a) All four queries succeed.
- (b) Only the first query succeeds.
- (c) Only the last query succeeds.

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```
1 happy(yolanda).  
2 listens2Music(mia).  
3 listens2Music(yolanda):- happy(yolanda).  
4 playsAirGuitar(mia):- listens2Music(mia).  
5 playsAirGuitar(yolanda):- listens2Music(yolanda).
```

Quiz

Why does `?- playsAirGuitar(yolanda).` succeed?

- (a) It is listed as a fact.
- (b) It reduces to `listens2Music(yolanda)`, which follows from `happy(yolanda)`.
- (c) Every happy person plays air guitar.

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```
1 happy(vincent).  
2 listens2Music(butch).  
3 playsAirGuitar(vincent):- happy(vincent),  
4     listens2Music(vincent).  
5 playsAirGuitar(butch):- happy(butch).  
6 playsAirGuitar(butch):- listens2Music(butch).
```

Quiz

For these queries, which option correctly describes the expected Prolog answers?

- (a) Both queries succeed.
- (b) Only the second query succeeds.
- (c) Only the first query succeeds.

```
?- playsAirGuitar(vincent).  
?- playsAirGuitar(butch).
```

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```
1  woman(mia).  
2  woman(jody).  
3  woman(yolanda).
```

Quiz

What answers does `?- woman(X).` return?

- (a) `X=mia; then X=jody; then X=yolanda.`
- (b) Only `X=mia.`
- (c) The query fails.

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References

```
1  woman(mia).  
2  woman(jody).  
3  loves(marsellus,mia).  
4  loves(pumpkin,honey_bunny).
```

Quiz

What does ?- loves(marsellus, X), woman(X). return?

- (a) X=mia.
- (b) X=pumpkin.
- (c) The query fails.

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```
1 loves(vincent,mia).
2 loves(marsellus,mia).
3 loves(pumpkin,honey_bunny).
4 loves(honey_bunny,pumpkin).
5
6 jealous(X,Y):- loves(X,Z), loves(Y,Z).
```

Quiz

According to this rule, what answers can

?- jealous(marsellus, W). return?

- (a) W=vincent only.
- (b) W=vincent; then W=marsellus.
- (c) The query fails.

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References

- A *search tree* shows how PROLOG tries to answer a query: successful and failed clause choices are both visible.
- A successful path ending in [] corresponds to a proof/answer. The whole tree also shows failed alternatives explored during backtracking.
- It makes execution explicit: *unification*, *rule expansion*, and *backtracking*.

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References

Follow this consistent procedure when drawing search trees:

1. Write the initial query in a box; treat it as a to-do list of goals.
2. Select the *leftmost* goal in the current box.
3. Try matching clauses for that predicate/arity from *top to bottom*; draw the branches from left to right in the same order.
4. Use a fresh copy of the selected clause each time it is tried.
5. If unification fails, mark the branch as `fail`.
6. If a fact succeeds, remove the selected goal and keep the remaining goals; write `[]` if no goals remain.
7. If a rule succeeds, replace the selected goal by the rule body and keep the remaining goals.

This procedure uses the PROLOG search order described in Learn Prolog Now! [Blackburn, Bos, and Striegnitz 2006]; the search-tree drawing notation follows Bos [Bos 2020].

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References

- In the examples below, `cl. n` means clause/line `n` in the numbered knowledge base.
- Each box is the current list of remaining goals; the **blue** goal is the next selected goal.
- We use `fail` for a dead branch and `[]` for a completed branch with no goals left.
- Substitutions such as `X=vincent` are written on edges or at successful leaves.
- Notation may vary, but the tree must consistently follow PROLOG's left-to-right and top-to-bottom search order.

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Example 1 (Search Tree)

Knowledge Base:

```
1 loves(vincent, mia).  
2 loves(marsellus, mia).  
3 jealous(A, B) :- loves(A, C), loves(B, C).
```

Query:

```
?- jealous(X, Y).
```

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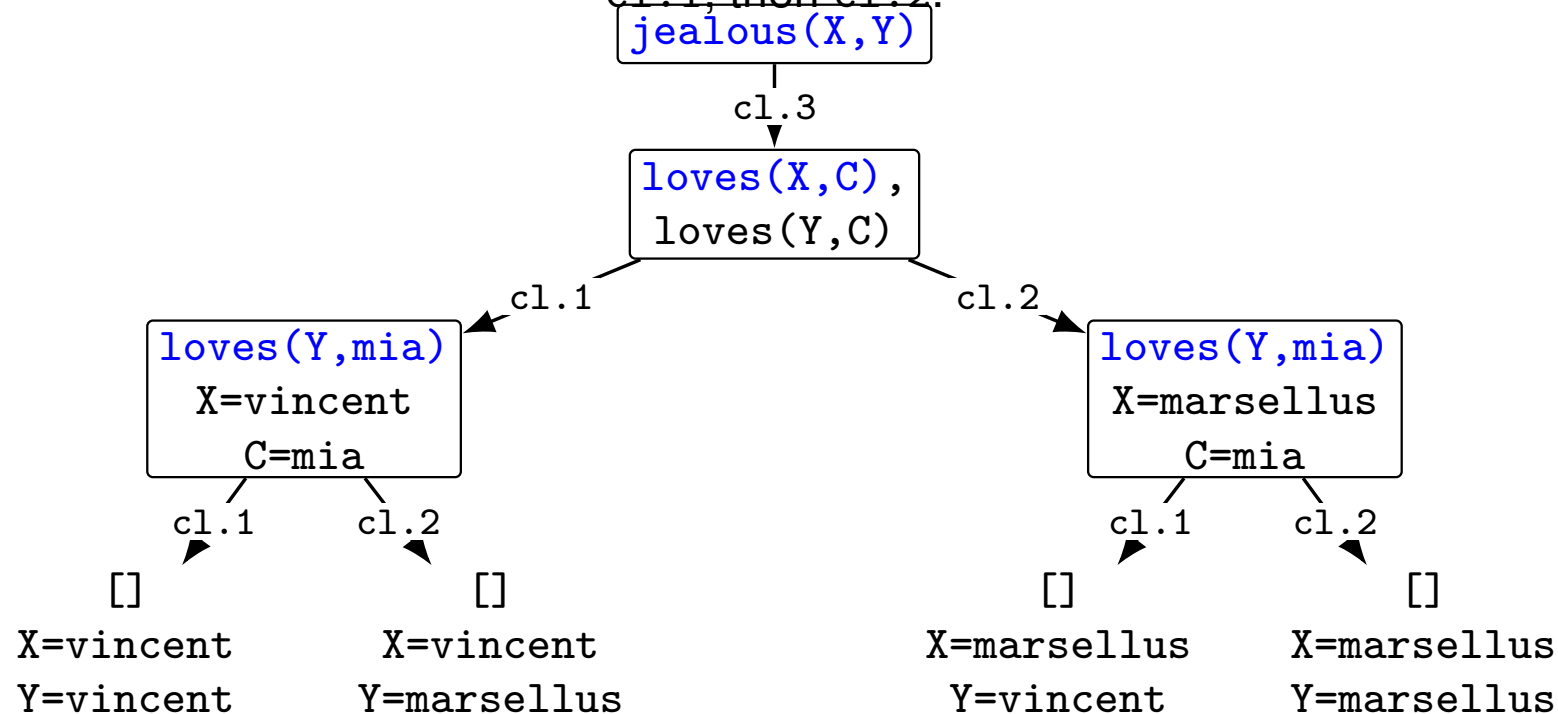
References

Search tree for the query ?- jealous(X, Y).

cl.1: loves(vincent,mia). cl.2: loves(marsellus,mia).

Step 1: expand jealous(X,Y) using loves(X,C). Step 2: try cl.1 for the leftmost goal loves(X,C). Step 3: solve the remaining goal loves(Y,mia) with cl.1.

Step 4: backtrack and try cl.2 for loves(Y,mia). Step 5: backtrack to loves(X,C) and try cl.2. Step 6: solve loves(Y,mia) on this branch with cl.1, then cl.2.



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Example 2 (Search Tree)

Knowledge Base:

```
1 loves(vincent,mia).  
2 loves(marsellus,mia).  
3 loves(pumpkin,honey_bunny).  
4 loves(honey_bunny,pumpkin).  
5 jealous(X,Y):- loves(X,Z), loves(Y,Z).
```

Query:

```
?- jealous(marsellus, W).
```

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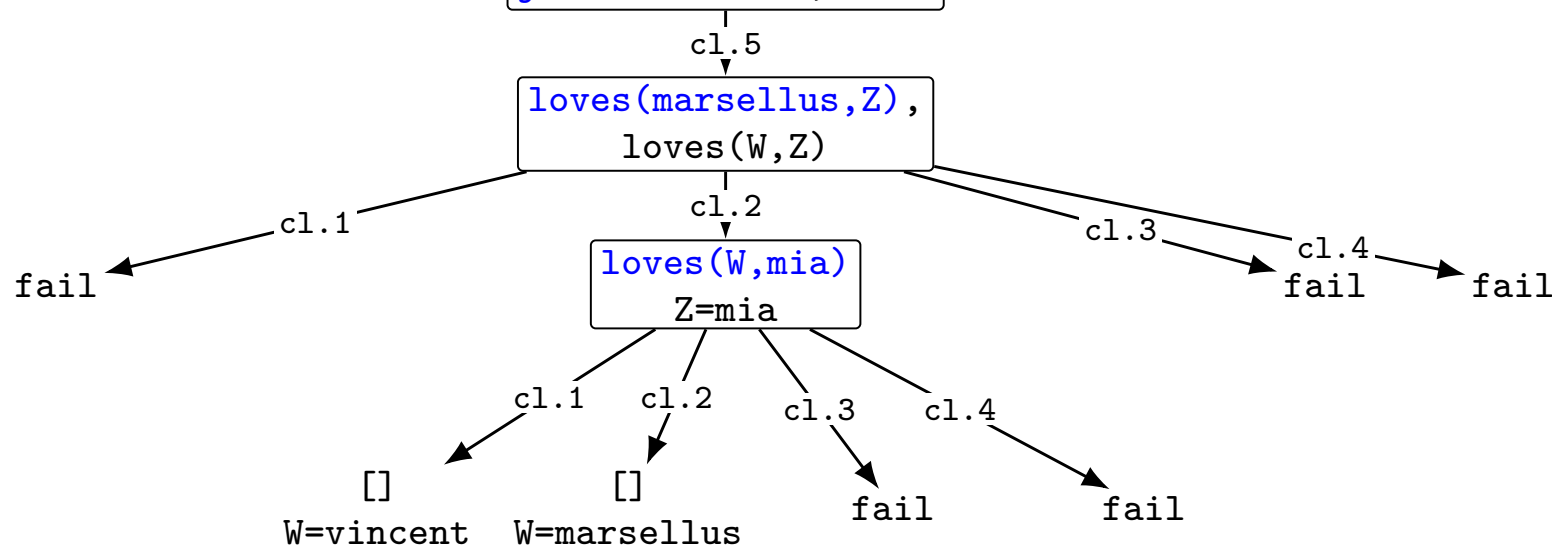
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References

Search tree for the query ?- jealous(marsellus, W).

cl.1: loves(vincent,mia). cl.2: loves(marsellus,mia).
cl.3: loves(pumpkin,honey_bunny). cl.4: loves(honey_bunny,pumpkin).
cl.5: jealous(X,Y) :- loves(X,Z), loves(Y,Z).
Step 1: expand jealous(marsellus,W) using cl.5. Step 2: try cl.1 for the leftmost goal loves(marsellus,Z); it fails. Step 3: try cl.2; this leaves the remaining goal loves(W,mia). Step 4: solve loves(W,mia) with cl.1. Step 5: backtrack and try cl.2 for loves(W,mia). Step 6: first try cl.3 and cl.4 for loves(W,mia); both fail. Then backtrack to loves(marsellus,Z) and try

cl.3 and cl.4 both fail.



Recursive Definitions

Introduction



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References

- In PROLOG, we can define predicates recursively.
- A recursive definition usually has:
 - At least one *base clause* (non-recursive)
 - At least one *recursive clause*
- For execution, the base clause must be *reachable* under PROLOG's clause order and subgoal order.
- We first use a small eating-and-digestion example, then return to Ertel's family-relationship example.

Recursive Definitions

Example: Eating



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References

Consider the following knowledge base:

```
1  is_digesting(X,Y)  :-  just_ate(X,Y).
2  is_digesting(X,Y)  :-
3      just_ate(X,Z),
4      is_digesting(Z,Y).
5
6  just_ate(mosquito,blood(john)).
7  just_ate(frog,mosquito).
8  just_ate(stork,frog).
```

- Line 1 is the *base clause*: direct eating is enough.
- Lines 2–4 are the *recursive clause*: one direct eating step, then the same question again.

Recursive Definitions

Declarative Meaning



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References

- *Declarative meaning*: The logical meaning of the PROLOG knowledge base
- Base clause (non-recursive):
 - “If x has just eaten Y, then X is now digesting Y”
- Recursive clause:
 - “If x has just eaten Z and Z is digesting Y, then x is digesting Y too”
- This captures the intuition of indirect digestion through food chains

Recursive Definitions

Procedural Meaning



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References

- *Procedural meaning*: How PROLOG actually executes the queries
- For a ground query such as `is_digesting(stork, mosquito)`, PROLOG:
 - First tries the base rule: “Has X just eaten Y?”
 - If that fails, tries the recursive rule by finding some Z where:
 - X has just eaten Z, AND
 - Z is digesting Y (recursive subgoal)
- For queries with variables, PROLOG may also *backtrack* after a success to look for more answers.

Recursive Definitions

Quiz: Answer Order



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```
1  is_digesting(X,Y) :- just_ate(X,Y).  
2  is_digesting(X,Y) :- just_ate(X,Z), is_digesting(Z,Y).  
3  just_ate(mosquito,blood(john)).  
4  just_ate(frog,mosquito).  
5  just_ate(stork,frog).
```

Quiz

What answers does `?- is_digesting(stork,Y).` return, in order?

- (a) `Y=frog; then Y=mosquito; then Y=blood(john).`
- (b) `Y=blood(john); then Y=mosquito; then Y=frog.`
- (c) Only `Y=frog.`

Recursive Definitions

The Importance of Base Cases



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Warning

A recursive definition needs a *reachable* stopping path.

Consider this dangerous rule:

```
p :- p.
```

- Declaratively: “If property p holds, then property p holds” (logical)
- Procedurally: Creates an *infinite loop*
 - To prove p , I need to prove p
 - To prove p , I need to prove p
 - ...and so on, forever
- A base clause helps only if PROLOG can actually reach it under its search order.

Recursive Definitions

Family Relationships: A Dangerous Symmetry Rule



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References

Consider the following knowledge base, adapted from Ertel's family-relationship example [Ertel 2025], Sec. 5.2.

```
1  child(oscar, karen, frank).
2  child(mary, karen, frank).
3  child(eve, anne, oscar).
4  child(henry, anne, oscar).
5  child(isabelle, anne, oscar).
6  child(clyde, mary, oscarb).
7
8  child(X,Z,Y) :- child(X,Y,Z).
9
10 descendant(X,Y) :- child(X,Y,Z).
11 descendant(X,Y) :- child(X,U,V), descendant(U,Y).
```

Now consider the query:

```
?- descendant(clyde,karen).
```

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Quiz: Nontermination



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Quiz

What happens with the query `?- descendant(clyde,karen).` in the program above?

- (a) PROLOG answers `false`.
- (b) PROLOG answers `true` immediately.
- (c) PROLOG follows an infinite branch before reaching the useful recursive descendant rule.

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Family Relationships: Where the Loop Comes From



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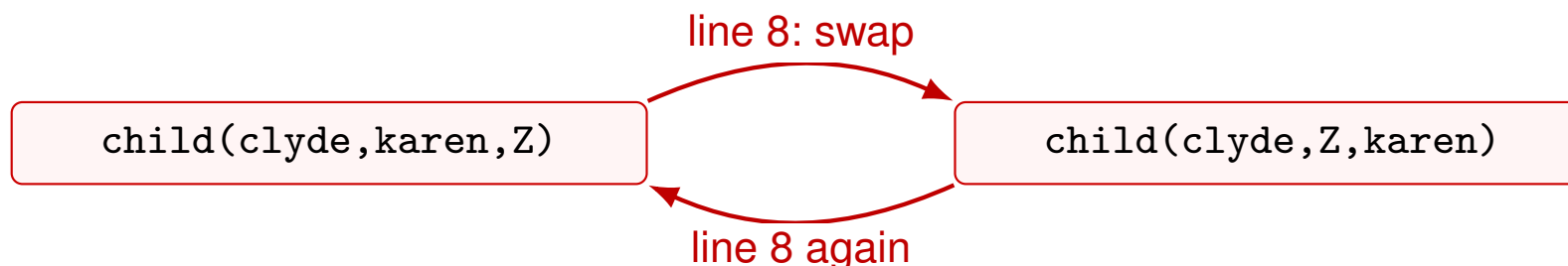
References

Important distinction

Not answered here means nontermination. It is *not* the answer false.

For `?- descendant(clyde,karen).`, PROLOG first tries line 10. If facts do not match the resulting `child` goal, line 8 can fire:

```
...                % omitted
8  child(X,Z,Y) :- child(X,Y,Z).  % swap
...                % omitted
10 descendant(X,Y) :- child(X,Y,Z). % base
...                % omitted
```



Recursive Definitions

Family Relationships: First Repair and Regression



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References

The following program is a first workaround for the nonterminating query.

rel01.pl

```
1  child(oscar, karen, frank).
2  child(mary, karen, frank).
3  child(eve, anne, oscar).
4  child(henry, anne, oscar).
5  child(isabelle, anne, oscar).
6  child(clyde, mary, oscarb).
7
8  descendant(X,Y) :- child(X,Y,Z).
9  descendant(X,Y) :- child(X,Z,Y).
10 descendant(X,Y) :- child(X,U,V), descendant(U,Y).
```

- The workaround: remove the recursive `child/3` symmetry rule; lines 8–9 test both parent positions directly in `descendant/2`.
- It answers `?- descendant(clyde,karen).` with `true`.
- Remaining problem: `?- child(eve,oscar,anne).` now fails, so this is only a *partial workaround*.

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Family Relationships: Separating Facts from Rules



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For this finite family example, both shown problems are avoided by separating stored facts from the derived predicate.

rel02.pl

```
1  child_fact(oscar, karen, frank).
2  child_fact(mary, karen, frank).
3  child_fact(eve, anne, oscar).
4  child_fact(henry, anne, oscar).
5  child_fact(isabelle, anne, oscar).
6  child_fact(clyde, mary, oscarb).
7
8  child(X,Z,Y) :- child_fact(X,Y,Z).
9  child(X,Z,Y) :- child_fact(X,Z,Y).
10
11 descendant(X,Y) :- child(X,Y,Z).
12 descendant(X,Y) :- child(X,U,V), descendant(U,Y).
```

- `child_fact/3` stores the original facts.
- `child/3` derives both parent orders *without recursively calling itself*.

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Quiz: Why `child_fact/3`?



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Quiz

Why does the program with `child_fact/3` avoid the previous `child/3` loop?

- (a) Stored facts are separated from the derived symmetric `child/3` predicate.
 - (b) Recursion is removed from `descendant/2`.
 - (c) PROLOG stops using depth-first search.
-
- This repair fixes only this loop: the old symmetric `child/3` rule kept recreating a `child/3` subgoal.
 - Other recursive rules can still loop, so a PROLOG programmer must *pay attention to processing* and *avoid infinite loops*.

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Note

As we have seen in the previous examples, *it is important to control the execution of PROLOG*.

- *Avoiding unnecessary backtracking* especially can lead to large increases in efficiency. One means to this end is the *cut* operator.
- The cut ! itself does *not* fail: when PROLOG reaches it, ! succeeds and commits to the choices already made.

```
p(X) :- q(X), !, r(X).  
p(X) :- s(X).
```

- If $q(X)$ fails, PROLOG never reaches !; it can still try the second clause $p(X) :- s(X)$.
- If $q(X)$ succeeds and PROLOG reaches !, it commits to the first clause. If $r(X)$ then fails, the second clause is not tried for this call. Thus the *position* of ! matters [Ertel 2025], Sec. 5.3; [Bratko 2011], Sec. 5.1.

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Example 3 (Cut operator in PROLOG)

max(X, Y, Max) means “the maximum of two numbers *X* and *Y* is *Max*”

max.pl

```
1 max(X,Y,Max) :-  
2     X >= Y,  
3     Max = X.  
4 max(X,Y,Max) :-  
5     X < Y,  
6     Max = Y.
```

maxwCut.pl

```
1 max(X,Y,Max) :-  
2     X >= Y,  
3     !,  
4     Max = X.  
5 max(_,Y,Y).
```

- Without cut.
- Query `?- max(3,2,Z), Z > 10.` first binds `Z=3` for `max(3,2,Z)`. Then `Z > 10` fails; backtracking reaches lines 4–6, where `3 < 2` fails.

- With cut.
- Query `?- max(3,2,Z), Z > 10.` reaches `!` and binds `Z=3`. When `Z > 10` fails, line 5 is not tried. The whole query fails.
- Cut can make programs harder to read.

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Quiz: Cut and Clause Heads



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Quiz

Consider this cut variant:

```
max(X,Y,X) :- X >= Y, !.  
max(_,Y,Y).  
?- max(3,2,2).
```

What answer does
PROLOG give?

- (a) It returns false.
- (b) It returns true.
- (c) It loops.

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Quiz: Cut and Backtracking



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Quiz

Consider this incorrect variant, where the cut is before the test:

```
max(X,Y,X) :- !, X >= Y.  
max(X,Y,Y).  
?- max(2,3,Z).
```

What happens?

- (a) Z=3, because PROLOG still tries clause 2.
- (b) false, because ! commits before $2 \geq 3$ is tested.
- (c) Z=2, because ! assigns X to Z.

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Predicate `fail`



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- The built-in predicate `fail` *always fails*.
- It is used to deliberately reject the current proof path.
- Common uses:
 - after printing one solution, to force PROLOG to backtrack and print the next one;
 - together with `cut`, to exclude a special case.
- See [Ertel 2025], Sec. 5.3; [Bratko 2011], Sec. 5.3.

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Example 4 (Predicate `fail` in PROLOG)

- In the family relationship example we can quite simply print out all children and their parents with the query

```
?- child_fact(X,Y,Z), write(X),  
  write(' is a child of '), write(Y),  
  write(' and '), write(Z), write('.'),  
  nl, fail.
```

- The corresponding output is

```
oscar is a child of karen and frank.  
mary is a child of karen and frank.  
eve is a child of anne and oscar.  
henry is a child of anne and oscar.  
isabelle is a child of anne and oscar.  
clyde is a child of mary and oscarb.  
false.
```

where the predicate `nl` causes a line break in the output. **What would be the output in the end without use of the `fail` predicate?**

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Quiz: What Does fail Do?



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Quiz

Consider this program:

```
snake(cobra).  
animal(cobra).  
  
likes(mary,X) :- snake(X), !, fail.  
likes(mary,X) :- animal(X).  
?- likes(mary,cobra).
```

What happens?

- (a) It succeeds because `animal(cobra)` is a fact.
- (b) It fails: cut commits to the snake case; `fail` rejects it.
- (c) It succeeds, then fails, because both rules are used.

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Example 5 (Negation as Failure)

- In the family relationship example, the query

```
?- child_fact(ulla,X,Y).
```

would result `false.` because there are no facts about `ulla`.

- This answer is *not logically correct*. Specifically, *it is not possible to prove that there is no object with the name ulla*. Here the prover E would correctly answer “No proof found.”
- Thus if *PROLOG answers false.*, this only means that *the query Q cannot be proved*. For this, however, $\neg Q$ *must not necessarily be proved*.

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- A collection of *ordered data*.
- Has zero or more elements enclosed by **square brackets** and **separated by commas** (',').

Example	Description
[A]	A list with one element
[]	An empty list
[34,tom,[2,3]]	A list with three elements where the third element is a list of two elements
[mia, love(honey), mia]	A list with three elements where the first and last elements are identical

- Like any object, a list can be unified with a variable

```
?- X = [Any, list, 'of elements'].  
X = [Any, list, 'of elements'].
```

Lists



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- A list can be decomposed into its *head* (first element) and *tail* (remaining elements) using the vertical bar operator ('|').
- For example, the list [A, B, C] can be decomposed as follows:

```
?- [Head|Tail] = [A, B, C].  
Head = A,  
Tail = [B, C].
```

- What are the head and the tail of the list [dead(z)]?
- **Note:** The empty list has neither a head nor a tail. That is, the empty list has no internal structure; for PROLOG, [] is a special, particularly simple, list.
 - What is the output of the following query?

```
?- [H|T] = [].
```

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Quiz: Splitting a List



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Quiz

What are the bindings for this query?

```
?- [X|Y] = [], dead(z), [2,[b,c]], [], Z].
```

- (a) $X=[], Y=[\text{dead}(z), [2, [b,c]], [], Z].$
- (b) $X=\text{dead}(z), Y=[[2, [b,c]], [], Z].$
- (c) The query fails.

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Quiz: Taking Two Elements



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Quiz

What are the bindings for this query?

```
?- [X,Y|W] = [], dead(z), [2,[b,c]], [], Z].
```

- (a) $X=[], Y=\text{dead}(z), W=[[2,[b,c]], [], Z].$
- (b) $X=[], Y=[\text{dead}(z), [2,[b,c]], [], Z], W=[].$
- (c) The query fails.

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Quiz: Taking a Longer Prefix



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Quiz

What are the bindings for this query?

```
?- [X1,X2,X3,X4|Tail] =  
    [], dead(z), [2,[b,c]], [], Z].
```

- (a) $X1=[], X2=\text{dead}(z), X3=[2, [b,c]], X4=[], \text{Tail}=[Z].$
- (b) $X1=[], X2=\text{dead}(z), X3=[2, [b,c]], X4=[], \text{Tail}=Z.$
- (c) The query fails.

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Quiz: Ignoring Positions



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Quiz

What are the bindings for this query?

```
?- [_ , X , _ , Y | _] =  
   [ [], dead(z), [2, [b,c]], [], Z] .
```

- (a) $X = [], Y = \text{dead}(z).$
- (b) $X = \text{dead}(z), Y = [].$
- (c) $X = [2, [b, c]], Y = Z.$

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Quiz: A Nested List



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Quiz

What is the binding for x?

```
?- [_ , _ , [_ | X] | _] =  
   [[ ] , dead(z) , [2 , [b , c]] , [ ] , Z] .
```

- (a) $X = [2, [b, c]]$.
- (b) $X = [[b, c]]$.
- (c) $X = [b, c]$.

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First recursive list program



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It's time to look at an example (from “Learn PROLOG Now!”) of a recursive PROLOG program for lists: the predicate `member/2`.

- Goal: given an object `X` and a list `L`, decide whether `X` belongs to `L`.
- The standard definition (one fact and one recursive rule):

```
1 member(X, [X|T]) .  
2 member(X, [H|T]) :- member(X, T) .
```

- First clause (fact): “`x` is a member of a list if `x` is the head of that list.” (uses the `|` operator)
- Second clause (recursive rule): “`x` is a member of a list if `x` is a member of the tail of the list.”
- Declaratively this is straightforward: the two clauses capture membership directly from the structure of lists.

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Procedural behaviour — examples



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Consider how PROLOG answers queries.

■ Immediate success:

```
?- member(yolanda, [yolanda, trudy, vincent, jules])
```

PROLOG succeeds immediately by unifying with the first clause.

■ Requires recursion/backtracking:

```
?- member(vincent, [yolanda, trudy, vincent, jules])
```

PROLOG tries the first clause (fails), uses the recursive clause repeatedly until the subgoal

```
member(vincent, [vincent, jules])
```

unifies with the first clause and succeeds.

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Failure and termination



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- If the queried element is not in the list, recursion eventually reaches the empty list and cannot proceed:

```
?- member(zed, [yolanda, trudy, vincent, jules]).
```

- PROLOG will derive successive goals

```
member(zed, [trudy, vincent, jules])  
member(zed, [vincent, jules])  
member(zed, [jules])  
member(zed, [])
```

and at `member(zed, [])`, neither clause applies (empty list cannot be split), so search stops and the answer is no.

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Enumerating members & a small improvement



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- `member/2` can be used with variables to enumerate elements:

```
?- member(X,[yolanda,trudy,vincent,jules]).  
X = yolanda ;  
X = trudy ;  
X = vincent ;  
X = jules ;  
no
```

- Small stylistic improvement: use anonymous variables for irrelevant parts

```
1 member(X,[X|_]).  
2 member(X,[_|T]) :- member(X,T).
```

Semantically identical, but clearer because each clause names only what matters.

Self-modifying Programs



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- PROLOG programs are not fully compiled, rather, they are interpreted by the WAM. Therefore *it is possible to modify programs at runtime*. A program can even modify itself.
- With commands such as **assert** and **retract**, facts and rules can be added to the knowledge base or taken out of it.
- Assert predicates
 - **assert(X)**: Adds a new fact or clause to the database. Term is asserted as the last fact or clause with the same key predicate.
 - **asserta(X)**: Same as **assert**, but adds a clause at the beginning of the database.
 - **assertz(X)**: Exactly same as **assert(X)**.
- Retract predicates
 - **retract(X)**: Removes fact or clause **x** from the database.
 - **retractall(X)**: Removes all facts or clauses from the database for which the head unifies with **x**.

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A simple application of **asserta** is the addition of derived facts to the beginning of the knowledge base with the goal of avoiding a repeated, potentially time-expensive derivation.

Example 6 (Family Relationship)

dynamic_rel.pl

```
1  child_fact(oscar, karen, frank).
2  child_fact(mary, karen, frank).
3  child_fact(eve, anne, oscar).
4  child_fact(henry, anne, oscar).
5  child_fact(isabelle, anne, oscar).
6  child_fact(clyde, mary, oscarb).
7
8  child(X,Z,Y) :- child_fact(X,Y,Z).
9  child(X,Z,Y) :- child_fact(X,Z,Y).
10
11 :- dynamic descendant/2.
12 descendant(X,Y) :- child(X,Y,Z), asserta(descendant(X,Y)).
13 descendant(X,Y) :- child(X,U,V), descendant(U,Y),
14                      asserta(descendant(X,Y)).
```

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```
?- [dynamic_rel].
true.

?- descendant(clyde, karen).
true .

?- listing(descendant).
:- dynamic descendant/2.

descendant(clyde, karen).
descendant(mary, karen).
descendant(X, Y) :-
    child(X, Y, Z),
    asserta(descendant(X, Y)).
descendant(X, Y) :-
    child(X, U, V),
    descendant(U, Y),
    asserta(descendant(X, Y)).

true.
```

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- By manipulating rules with `assert` and `retract`, even programs that change their own rules can be written. A related machine-learning idea is *genetic programming*, where programs or rules are changed automatically.
- In practice, however, it turns out that, due to the *huge number of senseless possible changes*, *changing the code by trial and error rarely leads to a performance increase*.
- *Systematic changing of rules*, on the other hand, *makes programming so much more complex* that, so far, such programs that extensively modify their own code have not been successful.
- *Machine learning* has been quite successful. However, only *very limited modifications* of the program code are being conducted here.

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Exercise 1

Understand how to solve this problem using PROLOG.

A farmer wants to bring a cabbage, a goat, and a wolf across a river, but his boat is so small that he can only take them across one at a time. The farmer thought it over and then said to himself: “If I first bring the wolf to the other side, then the goat will eat the cabbage. If I transport the cabbage first, then the goat will be eaten by the wolf. What should I do?”



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- The *programming of scheduling systems*, in which many (sometimes complex) logical and numerical conditions must be fulfilled, *can be very expensive and difficult with conventional programming languages*.
- This is precisely where *logic could be useful*.
- An approach is to simply *write all logical conditions in PL1 and then enter a query*. *Usually this direct PL1 approach is not convenient*. As the penguin example showed, PL1 does not guess facts that we did not write down. For example, from `penguin(tweety)` it does not guess that Tweety is not a raven; such exclusions must be stated explicitly.
- For scheduling problems, the relevant restrictions can be stated explicitly as constraints. *Constraint Logic Programming (CLP)* [Jaffar and Lassez 1987], which allows the *explicit formulation of constraints for variables*, offers an elegant and efficient mechanism for such constraint-heavy search problems.
 - The *interpreter constantly monitors the execution of the program for adherence to all of its constraints*.
 - The programmer is fully relieved of the task of controlling the constraints, which in many cases can greatly simplify programming.

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Example 7 (Applying the CLP mechanism of GNU Prolog (The finite domain (FD) constraint solver))

The secretary of Albert Einstein High School has to come up with a plan for allocating rooms for final exams. He has the following information: the four teachers Mayer, Hoover, Miller and Smith give tests for the subjects German, English, Math, and Physics in the ascendingly numbered rooms 1, 2, 3 and 4. Every teacher gives a test for exactly one subject in exactly one room. Besides that, he knows the following about the teachers and their subjects.

- (1) Mr. Mayer never tests in room 4.
- (2) Mr. Miller always tests German.
- (3) Mr. Smith and Mr. Miller do not give tests in neighboring rooms.
- (4) Mrs. Hoover tests Mathematics.
- (5) Physics is always tested in room number 4.
- (6) German and English are not tested in room 1.

Who gives a test in which room?

Constraint Logic Programming



raumplan.pl

```
1  %%% Run in GNU Prolog
2  start :-
3      fd_domain([Mayer, Hoover, Miller, Smith],1,4),
4      fd_all_different([Mayer, Miller, Hoover, Smith]),
5
6      fd_domain([German, English, Math, Physics],1,4),
7      fd_all_different([German, English, Math, Physics]),
8
9      fd_labeling([Mayer, Hoover, Miller, Smith]),
10
11     Mayer #\=4,                % Mayer not in room 4
12     Miller #= German,          % Miller tests German
13     dist(Miller, Smith) #>= 2, % Distance Miller/Smith >= 2
14     Hoover #= Math,            % Hoover tests mathematics
15     Physics #= 4,              % Physics in room 4
16     German #\= 1,              % German not in room 1
17     English #\= 1,            % English not in room 1
18     nl,
19     write([Mayer, Hoover, Miller, Smith]), nl,
20     write([German, English, Math, Physics]), nl.
```

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GNU Prolog predicates



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■ GNU Prolog built-in predicates:

- `fd_domain(Vars, Lower, Upper)` constrains each element `X` of `Vars` to take a value in `Lower..Upper`.
- `fd_all_different(List)` constrains all variables in `List` to take distinct values.
- `fd_labeling(Vars, Options)` assigns a value to each variable `X` of the list `Vars` according to the list of labeling options given by `Options`. This predicate is re-executable on backtracking. `fd_labeling(Vars)` is equivalent to `fd_labeling(Vars, [])`.

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Reading the program



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- The variables Mayer, Hoover, Miller, Smith as well as German, English, Math, Physics can each take on an integer value from 1 to 4 as the room number. (Lines 3–6.)
- A binding Mayer = 1 and German = 1 means that Mr. Mayer gives the German test in room 1.
- Lines 4 and 7 ensure that the four particular variables take on different values.
- Line 9 starts the finite-domain search: GNU Prolog chooses concrete room numbers for the teacher variables. If a later constraint fails, PROLOG backtracks to line 9 and tries another assignment. The subject rooms are then determined by the constraints below. Without this labeling step, GNU Prolog may print finite-domain intervals instead of concrete room numbers.
- In lines 11–17 the constraints are given, and the remaining lines output the room numbers for all teachers and all subjects in a simple format.

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The program is loaded into GNU Prolog with
`['raumplan.pl']`., and with `start.` we obtain the output

```
[3,1,2,4]
```

```
[2,3,1,4]
```

```
true ?
```

```
yes
```

This output corresponds to the plan

Room num.	1	2	3	4
Teacher	Hoover	Miller	Mayer	Smith
Subject	Math	German	English	Physics

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